THE DEGREE OF A PULLED-BACK LOCALLY FREE MODULE ALONG A NONCONSTANT MORPHISM OF PROPER CURVES IS MULTIPLIED BY THE DEGREE OF THE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Definition of the degree of a dominant morphism of proper curves. Stacks Project, Morphisms, Definition morphisms-definition-degree (tag 02NY). Supplies the meaning of deg(X/Y) used throughout the computation.
- Relative affine cohomology, the projection formula and the rank formula for the degree. Stacks Project, Cohomology of Schemes, Lemma coherent-lemma-relative-affine-cohomology (tag 089W); Cohomology, Lemma cohomology-lemma-projection-formula (tag 01E8); Varieties, Lemma lemma-degree-on-proper-curve (tag 0AYV). Provide the three equalities of the displayed chain.

Source reference. Original reference 02NY: sources/stacks-chow/tags.txt, source lines 3497–3497. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 089W: sources/stacks-chow/tags.txt, source lines 10511–10511. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01E8: sources/stacks-chow/tags.txt, source lines 1825–1825. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0AYV: sources/stacks-chow/tags.txt, source lines 13803–13803. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let $f : X \to Y$ be a nonconstant morphism of proper curves over k. Let $\mathcal{E}$ be a locally free $\mathcal{O}_Y$-module. Then*

$$\deg(f^*\mathcal{E}) = \deg(X/Y)\deg(\mathcal{E})$$

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Proof. The degree of X over Y is defined in Morphisms, Definition 02NY. Thus $f_*\mathcal{O}_X$ is a coherent $\mathcal{O}_Y$-module of rank $\deg(X/Y)$, i.e., $\deg(X/Y) = \dim_{\kappa(\xi)}(f_*\mathcal{O}_X)_\xi$ where ξ is the generic point of Y. Thus we obtain

$$\begin{aligned}\chi(X, f^*\mathcal{E}) &= \chi(Y, f_*f^*\mathcal{E}) \\ &= \chi(Y, \mathcal{E} \otimes f_*\mathcal{O}_X) \\ &= \deg(X/Y)\deg(\mathcal{E}) + n\chi(Y, f_*\mathcal{O}_X) \\ &= \deg(X/Y)\deg(\mathcal{E}) + n\chi(X, \mathcal{O}_X)\end{aligned}$$

as desired. The first equality as f is finite, see Cohomology of Schemes, Lemma 089W. The second equality by projection formula, see Cohomology, Lemma 01E8. The third equality by Lemma 0AYV. □